

Algorithm 5: BIC-cherry + expand via safe-pairs (strict only)**Require:** properly coloured digraph (G, σ) on L with the sicor-in-hub property**Ensure:** leaf-coloured network (N, σ)

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1:  $V(N) \leftarrow L \cup \{\rho\}$ ;  $E(N) \leftarrow \emptyset$ 
2: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ BIC-cherry network
3:   add  $p_{xy}$  with edges  $(\rho, p_{xy}), (p_{xy}, x), (p_{xy}, y)$ 
4: end for
5: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ expansion
6:   if  $(x, y) \notin E(G)$  and  $(y, x) \notin E(G)$  then ▷ both directions missing
7:      $(x', y') \leftarrow \text{FINDJOINTCANDIDATES}(G, x, y)$ 
8:   else ▷ at most one direction missing
9:     for all  $(a, b) \in ((x, y), (y, x))$  with  $(a, b) \notin E(G)$  do  $b' \leftarrow$   

        $\text{FINDCANDIDATE}(G, a, b)$ 
10:   end if
11:   for all  $(a, b) \in ((x, y), (y, x))$  with  $(a, b) \notin E(G)$  do
12:     add new vertex  $q$  with edges  $(p_{xy}, q), (q, a), (q, b')$  ▷  $[ab : ab']$ 
13:     remove edge  $(p_{xy}, a)$  ▷ shortcut
14:   end for
15: end for
16: return  $(N, \sigma)$ 

17: function  $\text{FINDCANDIDATE}(G, a, b)$ 
18:    $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G)\}$ 
19:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b), (v, a) \in E(G)\}$ 
20:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b)\}$ 
21:   return random element of  $C$ 
22: end function

23: function  $\text{FINDJOINTCANDIDATES}(G, x, y)$ 
24:    $X_1 \leftarrow \{v \in L : \sigma(v) = \sigma(x), (y, v) \in E(G)\}$ 
25:    $Y_1 \leftarrow \{v \in L : \sigma(v) = \sigma(y), (x, v) \in E(G)\}$ 
26:    $S \leftarrow \{(x', y') \in X_1 \times Y_1 : (x', y') \notin E(G), (y', x') \notin E(G)\}$ 
27:   if  $S \neq \emptyset$  then return random element of  $S$ 
28:    $X_2 \leftarrow \{v \in L \setminus \{x\} : \sigma(v) = \sigma(x)\}$ 
29:    $Y_2 \leftarrow \{v \in L \setminus \{y\} : \sigma(v) = \sigma(y)\}$ 
30:    $S \leftarrow \{(x', y') \in X_2 \times Y_2 : (x', y') \notin E(G), (y', x') \notin E(G)\}$ 
31:   if  $S \neq \emptyset$  then return random element of  $S$ 
32:   if  $X_1 \times Y_1 \neq \emptyset$  then return random element of  $X_1 \times Y_1$ 
33:   return random element of  $X_2 \times Y_2$  ▷ no safe pair exists
34: end function

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